BRAC UNIVERSITY
Inspiring Excellence

CSE490 (D)

INTRODUCTION TO FOOTBALL DATA AND ANALYTICS

IN ASSOCIATION WITH BANGLADESH FOOTBALL FEDERAITON

SPRING 2026

- Professional Certificate from BFF
- Internship Opportunity at BFF Data Science Department after successful completion
- Learning from the industry experts of Football Data Science
- Opportunity to co-author research papers for Sports Data Conferences

CSE490 (D)
Introduction to
Football Data & Analytics

Lecture-3

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Why Modeling

- We want to predict what is going to happen
- Modeling captures our understanding of football
- Can you guess the outcome from this picture?

Linear Regression

- Simple, explainable way to find relationships in data

$$Y = \underbrace{\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_k}_{f(X;\beta)} + \epsilon$$

here the (output) variable Y is a found as combination of k input variables plus some noise/error $\in$.

- The simplest objective function for this type of regression is the minimization of the sum of squared residuals (SSR). Other losses such as mean squared error (MSE), mean absolute error (MAE) etc. can also be used.
- Linear regression using SSR as the loss function is also called ordinary least square (OLS) regression. Scikit-Learn's implementation of linear regression is also an OLS regression.

Linear Regression

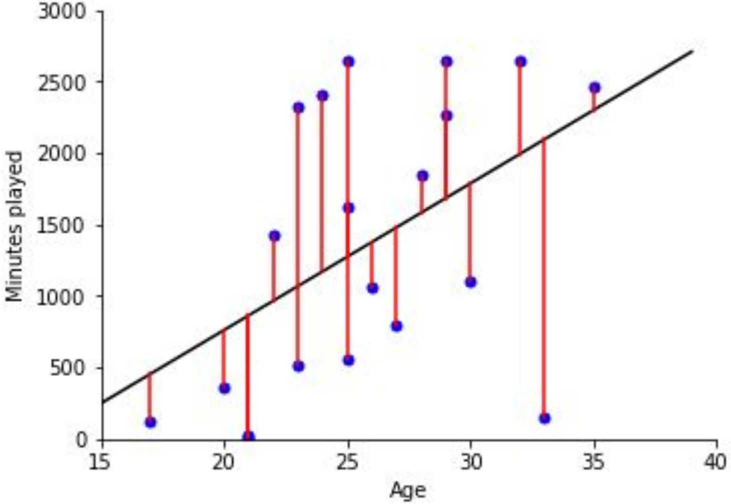
- SSR is a total measure of error (depends on dataset size).
- MSE is an average measure of error (normalized per observation).
- SSR is used in deriving OLS, whereas MSE is used to evaluate and compare model performance.

$$SSR = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

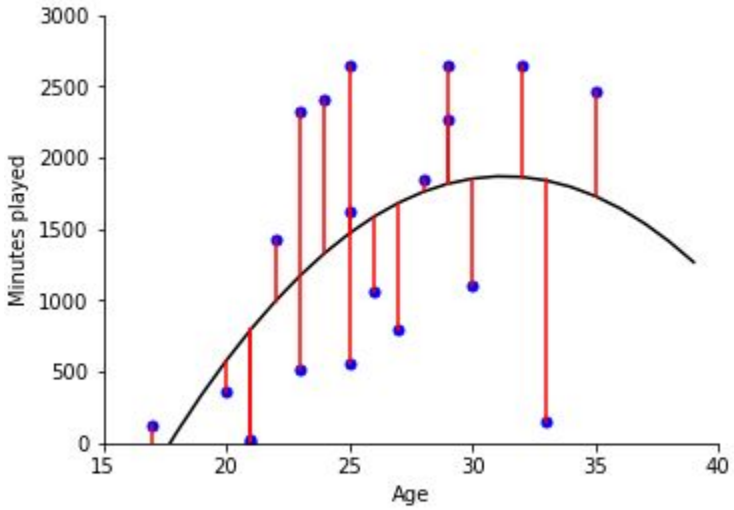
Linear Regression

$$\text{Minutes Played} = \beta_0 + \beta_1 \text{Age}$$



Linear Regression

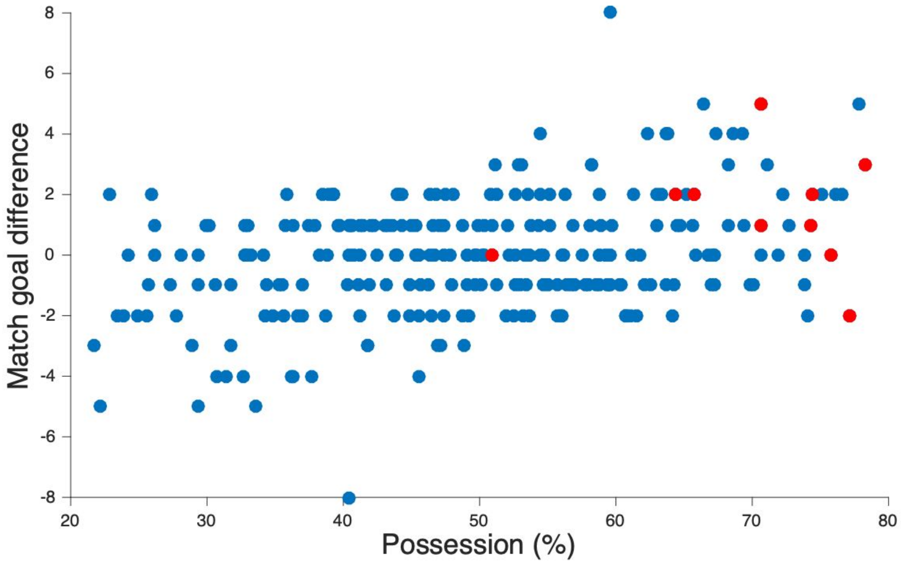
$$\text{Minutes Played} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{Age}^2$$



Self Read: How regression model is trained

Confusion

- Does statistics tell us nothing?!
- Statistics never told you that more possession means more goals, you assumed that!



Logistic Regression

- Used when the output variable Y is binary (0/1 outcome)

A logistic regression model has the form

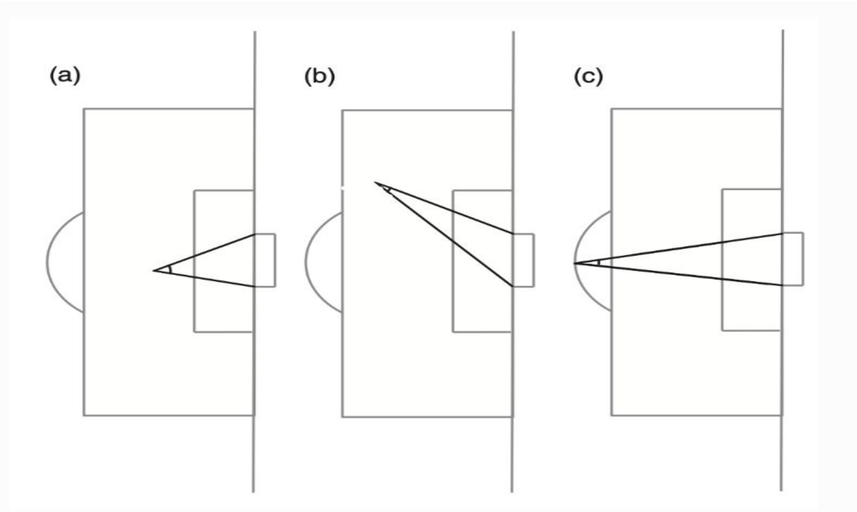
$$p(Y=1 \mid x_1,\ldots,x_n) = \frac{1}{1+\exp(-(\beta_0+\beta_1 x_1 + .. + \beta_n x_n))}$$

where $x_1,\ldots,x_n$ is a list of features and Y is the event to be predicted (1 means it happens, 0 means it doesn't). So, $p(Y=1 \mid x_1,\ldots,x_n)$ means the probability event Y occurs given features x_i.

- The objective function is typically the maximum likelihood estimation (MLE), minimizing the log-loss (cross-entropy).
- MLE finds the curve that gives high probability to what actually happened.

Expected Goals (xG)

- Which shot is most likely to be a goal?
- measure the quality of a chance created.
- When we say that a shot had an xG of 0.2 (or 20%), we mean that typically shots of this type will be scored 20% of the time
- Which features do you think affect expected goals?



Expected Goals (xG)

- **Shot Location**
  – Distance to goal
  – Angle to goal

- **Body Part Used**
  – Strong/weak foot
  – Header vs foot

- **Type of Assist**
  – Through ball, cross, cutback, rebound

- **Defensive & Goalkeeper Context**
  – Defensive pressure
  – Number/position of defenders
  – Goalkeeper positioning

- **Game Situation**
  – Open play vs set piece
  – Counterattack
  – Penalty / one-on-one

Expected Goals (xG)

- Do you think the team with the highest xG is the worthy winner ?
- xG tells us little more about the scoreline than the result itself.
- Take, for instance, a hard-fought match in the Champions League qualifiers between Bayern Munich and Lyon. Bayern won 1-0, but recorded 0.45-1.68 in xG.

Logistic Regression

- Which shot is most likely to be a goal?